Task 2: Descriptive Statistics & Exploratory Analysis

Short report | Illustrative customer-purchasing dataset (200 records)

Dataset and approach

Because no source dataset was attached, this report uses a reproducible illustrative dataset of 200 customers. It includes age, city, purchase frequency, average order value, and estimated total spend. The figures below demonstrate the analysis workflow and should not be treated as findings about a real business.

Key summary statistics

Metric	Mean	Median	Minimum	Maximum
Age (years)	37.6	38.0	18	71
Purchases per customer	5.04	5.0	1	22
Average order value (INR)	■1,907	■1,837	■675	■9,200
Total spend (INR)	■9,496	■8,240	■1,308	■55,200

Trends and relationships

• Purchase frequency and total spend are strongly positively related (Pearson r = 0.75); average order value and total spend are also positively related (r = 0.57). This is expected because total spend is calculated as purchase frequency × average order value. • City-level totals are not evenly distributed in this sample. The highest total-spend city is Bengaluru (■426,231 across 42 customers). • The age group with the highest average spend per customer is 46–55, at approximately ■10,944.

Outliers and unusual patterns

Using the 1.5×IQR rule, 3 customers are flagged for unusually high or low average order values, and 4 are flagged for purchase frequency. The sample intentionally includes a few unusually large order values and high purchase counts to illustrate detection. Outliers should be checked against source records before removal: they may reflect genuine high-value customers, data-entry errors, or special events.

Business questions and recommendations

1. Which customers may be most valuable? Prioritize customers with both high purchase frequency and high order value; test loyalty benefits or VIP offers with this group.
2. Where should marketing focus? Compare city-level spend with customer counts and campaign costs before shifting budget; a larger total may simply reflect more customers.
3. How should unusual records be handled? Validate extreme order values and purchase counts, then document whether each is retained, corrected, or excluded.
4. What should be explored next? Add transaction dates, product categories, acquisition channel, and margin to distinguish seasonality, product mix, and profitable revenue from spend alone.

Conclusion

Descriptive statistics quickly summarize customer profiles, spending distributions, and possible anomalies. For reliable business decisions, repeat this workflow on the actual dataset and verify the data definitions, missing values, duplicate records, and outlier causes first.